

PUBLISHED

Question	Answer	Marks
3(a)(i)	any three from: 1 glucose diffuses into the cell (through, A / transport protein) ; 2 <i>idea that</i> the enzyme / B, loses phosphate group ; 3 (dephosphorylated) B binds to lactose permease ; 4 causes, shape change / allosteric change, to lactose permease ; 5 lactose permease cannot function (so lactose cannot enter) ;	3
3(a)(ii)	any two from: 1 no / less, (protein) synthesis of, lactose permease / β-galactosidase / transacetylase or no / less, transcription / expression, of (named structural) genes (of lac operon) ; 2 no / less, ATP / energy, wasted ; 3 no / fewer, amino acids wasted ; 4 glucose is respired more easily / AW or fewer steps / less energy, required to break down glucose or lactose has to be broken down (to glucose and galactose) ;	2